

The Evolution of Random Graphs: Phase Transitions and Connectivity

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Abstract

This report provides a rigorous mathematical analysis of the Erdős-Rényi random graph model. We investigate the sizes of the connected components and how they evolve for a graph on n vertices as edges are uniformly and randomly added. We use DFS based approach and Chernoff bounds to show the emergence of a giant component as probability of selection of edge cross a certain threshold, and observe when the graph gets fully connected.

1 Introduction and the Random Graph Model

A random graph is not a single, fixed graph, but rather a probability space over a family of graphs.

We begin by defining two ways we could treat random graphs: the uniform model $G(n, m)$ and the binomial model $G(n, p)$.

1.1 The $G(n, m)$ and $G(n, p)$ Models

Definition 1.1 (The $G(n, m)$ Model). Let n be a positive integer and m be an integer such that $0 \leq m \leq \binom{n}{2}$. The random graph $G(n, m)$ is a probability space over the set of all graphs on the vertex set $V = \{1, 2, \dots, n\}$ containing exactly m edges. Any such graph is chosen uniformly at random.

Definition 1.2 (The $G(n, p)$ Model). Let n be a positive integer and p a real number such that $0 \leq p \leq 1$. The random graph $G(n, p)$ is a probability space over all graphs on the vertex set $V = \{1, 2, \dots, n\}$ determined by the rule that each of the $\binom{n}{2}$ potential edges is included with probability p independently.

We shall see that using $G(n, p)$ model is much easier to use in our proofs than $G(n, m)$. Since the expected number of edges in $G(n, p)$ is $p\binom{n}{2}$, we can say that the an $G(n, p')$ model “equivalent” model to $G(n, m)$ is where

$$p' \sim \frac{m}{\binom{n}{2}} \approx \frac{2m}{n^2}$$

so that expected number of edges is m in our model $G(n, p')$. Also note that:

- Adding $m = n/2$ edges corresponds to an edge probability of $p \approx \frac{1}{n}$.
- Adding $m = 2n$ edges corresponds to an edge probability of $p \approx \frac{4}{n}$.

Because the behavior of the two models is similar, ¹, we will conduct our rigorous proofs exclusively using the $G(n, p)$ probability space. Just for completeness we define the following as well.

Definition 1.3 (With High Probability). A graph-theoretic property A is said to hold **with high probability (whp)** if the probability that A holds tends to 1 as the number of vertices n approaches infinity.

1.2 The Erdős-Rényi Phase Transition

Before getting into all the details, we feel it is imperative to give a sort of feel how the “random graph” $G(n, p)$ grows as we increase probability of selection of an edge p . Indeed, the graph undergoes a sudden, dramatic shift around the probability $p = 1/n$.

We can formalize this behavior by examining the graph slightly below and slightly above this threshold, separated by a small constant ϵ .

Theorem 1.1 (The Phase Transition). *Let $\epsilon > 0$ be a small enough constant, and let $G \sim G(n, p)$.*

1. **The Subcritical Phase** ($p \leq \frac{1-\epsilon}{n}$): *In this region, the graph is highly “fragmented” or “disconnected”. All connected components of G are tiny, with the maximum size being strictly bounded by $O(\ln n)$ (specifically, at most $\frac{7}{\epsilon^2} \ln n$) with **whp**.*
2. **The Supercritical Phase** ($p \geq \frac{1+\epsilon}{n}$): *By crossing the $1/n$ threshold, the graph undergoes a sudden “crystallization”. Smaller components join to form bigger components. G contains a unique “Giant Component” consisting of at least $\frac{\epsilon n}{2}$ vertices **whp**, representing a linear fraction $\Theta(n)$ of the entire graph.*

¹for proof, see bollobas’s random graph book

The Critical Window: The boundary around $p = 1/n$ acts much like the freezing point of water. The transition from tiny $O(\ln n)$ trees to a massive $\Theta(n)$ giant component does not happen instantly, but rather inside a very small range around $1/n$ known as the *critical window*.

This window occurs when $p = \frac{1}{n} \pm \frac{c}{n^{4/3}}$ (or equivalently, when our parameter $\epsilon = \Theta(n^{-1/3})$). Inside this exact boundary, the graph is in a highly volatile state. The connected components are neither purely logarithmic nor linearly massive; instead, the largest component scales as exactly $\Theta(n^{2/3})$.

2 Depth-First Search (DFS)

We are going to analyze everything the way it was done in [reference]. For a graph $G = (V, E)$, we know that DFS algorithm maintains three sets of vertices: S, U, T . Here, S denotes set of vertices for which we have completed their dfs call, U denoting the sequence of vertices which are currently in stack, and the rest of the unvisited vertices lies in $T = V - (S \cup U)$. We start off with S, U being empty and T being V and we stop when U and T both becomes empty.

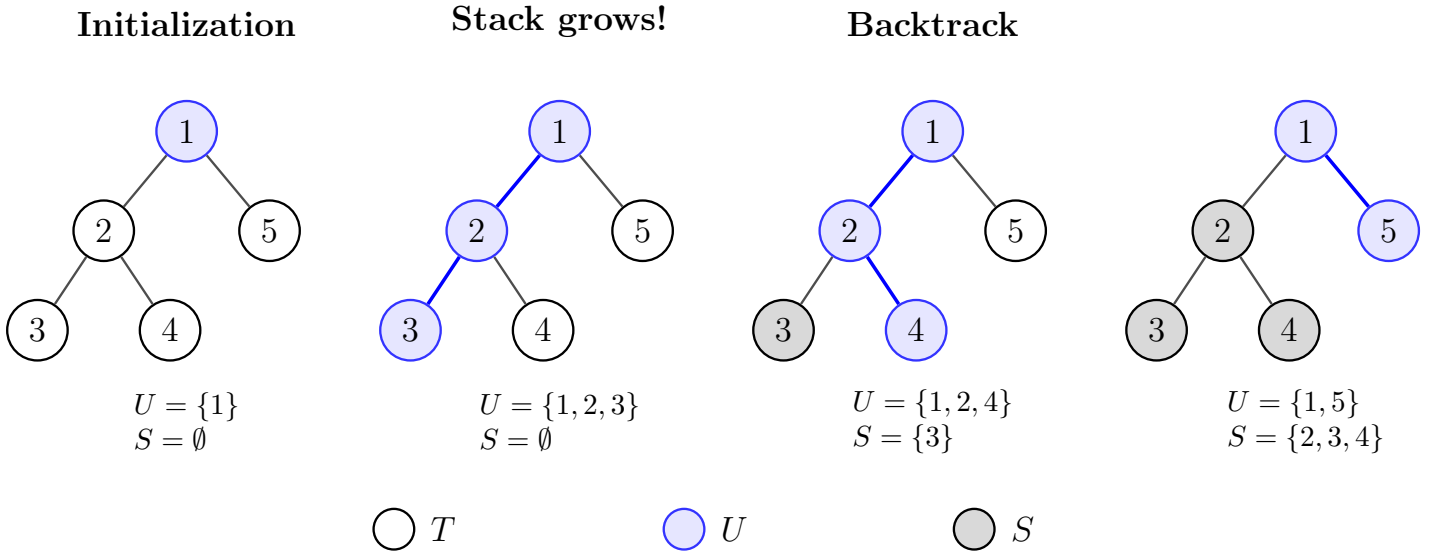


Figure 1: A step-by-step visualization of the Depth-First Search (DFS) state transitions.

Note that at each step:

- If U is not empty, then the algorithm looks for neighbors of the last vertex v added in U in T , (moves it from $T \rightarrow U$), otherwise if no neighbors of v lies in T , vertex v is moved from $U \rightarrow S$.
- Otherwise, the algorithm looks for the first vertex in T and adds it to U , while deleting it from T itself.

Here, WLOG we assumed that querying vertices from T follows in strictly increasing fashion (smaller labeled vertices gets queried first), we shall call this order σ from now on. Note that when first vertex of a component C gets into U (U is empty beforehand) then the algorithm starts revealing the connected component C and finishes when U becomes empty again. We call this time when the algorithm starts exploring a component and finishes it up an *epoch*.

Now before we begin, let us setup some definitions. Define a random variable X_t as follows: at a time t , if we are querying some edge according to σ and if it exists, $X_t = 1$ and 0 otherwise. We shall prove following two properties which will be used later repeatedly:

Property 1: $|S \cup U| \geq \sum_{i=1}^t X_i$: Note that while inside some component C , at $t = t + 1$ either some v moves from $U \rightarrow S$ then both sides of the equation don't change (as $X_{t+1} = 0$, we couldn't find any neighbour of the last vertex of v in T), or some vertex v moves from $T \rightarrow U$ then both sides increase by 1. Now if C becomes empty, we add some v from T (if still non-empty) to C increasing the left side by 1 while not needing to perform any query on edges. This finishes the proof.

Property 2: $|U| \leq 1 + \sum_{i=1}^t X_i$: This holds trivially as whenever X_t is 1, $|U|$ must increase (since a new vertex has been added from $T \rightarrow U$), and at any time t , $|U|$ can't increase without X_t being 1. This finishes the proof.

3 Probabilistic Preliminaries

We shall be needing some famous probabilistic inequalities, so let's write them up now:

Theorem 3.1 (Chebyshev's Inequality). *Let X be a random variable with finite expected value μ and finite non-zero variance σ^2 . For any real number $t > 0$, the probability that X deviates from its mean by at least t is bounded by:*

$$\Pr(|X - \mu| \geq t) \leq \frac{\sigma^2}{t^2}$$

Theorem 3.2 (Chernoff Bounds). *Let $X_1, X_2, \dots, X_m$ be independent Bernoulli random variables. Let $S = \sum_{i=1}^m X_i$ be their sum, and let $\mu = \mathbb{E}[S]$ be the expected value.*

1. *For any $t > 0$ such that $t < \mu$:*

$$\Pr(|S - \mu| \geq t) \leq 2 \exp\left(-\frac{t^2}{3\mu}\right)$$

2. *For any fractional deviation $\delta \geq 1$:*

$$\Pr(S \geq (1 + \delta)\mu) \leq \exp\left(-\frac{\delta^2 \mu}{3}\right)$$

We omit their proofs as they can be found online (or even in lecture notes of our course).

4 A bit to the left... $p \leq \frac{1-\epsilon}{n}$

We first analyze the graph in the left of the critical region, where the edge probability is slightly below the critical threshold. We will rigorously prove that all the connected components will be logarithmic in size, and establish what we require to show this along the way.

Theorem 4.1. *Let $\epsilon > 0$ be a sufficiently small constant, and let $G \sim G(n, p)$ with $p = \frac{1-\epsilon}{n}$. Then **whp**, the size of the largest connected component in G is strictly bounded by $k = \frac{7}{\epsilon^2} \ln n$.*

Let's proceed by contradiction. Assume that G contains at least one connected component C with strictly more than k vertices. We run the DFS algorithm on G and observe the specific tie during which component C is discovered and explored.

Consider the exact moment during this run when the DFS algorithm successfully discovers the $(k+1)$ -th vertex of C . At this moment, the algorithm is about to push this new vertex onto the stack U . Now where do these k vertices that we have discovered in C lie? They are precisely either in current stack or their exploration is completed. Define $\Delta S = S \cup C$ (explored vertices but lying in the current component), then $|\Delta S \cup U| = k$.

Also note that since every newly discovered vertex corresponds to exactly one successful edge query, the DFS algorithm must have received exactly k positive answers ($X_i = 1$) from our sequence of Bernoulli random variables during this epoch.

Let us calculate the absolute maximum number of queries the algorithm could have made in this epoch to achieve these k successes. The algorithm could only have made queried those edges that are touching the k currently discovered vertices. There are two types of such edges:

1. **Internal Queries:** Edges between two vertices that are both already discovered within the component. The absolute maximum number of such pairs is $\binom{k}{2}$.
2. **Boundary Queries:** Edges between a discovered vertex and an undiscovered vertex in the remainder of the graph. The absolute maximum number of such pairs is $k(n-k)$.

Adding these together provides a strict upper bound on the total queries t made by the algorithm during this epoch:

$$t \leq \binom{k}{2} + k(n-k)$$

$$t \leq \frac{k^2 - k}{2} + kn - k^2 = kn - \frac{k^2 + k}{2} < kn$$

Now we ask ourselves: is this even possible? Can we get at least k successes in a span of kn consecutive bournoulli trials. We show that this is *almost* impossible.

Claim: The probability of finding k successes in any kn consecutive queries is *very small*

Let I be a fixed interval of kn queries. The total number of successes S_I in this interval follows a Binomial distribution $B(kn, p)$. The expected number of successes is:

$$\mu = knp = (kn) \left(\frac{1-\epsilon}{n} \right) = k(1-\epsilon)$$

We must bound the probability that $S_I \geq k$. We can express our k as:

$$k = (1+\delta)\mu \implies k = (1+\delta)k(1-\epsilon) \implies 1 = (1+\delta)(1-\epsilon)$$

Solving for δ yields $\delta = \frac{\epsilon}{1-\epsilon}$. Because ϵ is small, and so is δ , we apply the standard Chernoff upper tail bound $\Pr(S_I \geq (1+\delta)\mu) \leq \exp\left(-\frac{\delta^2\mu}{3}\right)$.

Indeed (subtituting the value of k too),

$$\Pr(S_I \geq k) \leq \exp\left(-\frac{k(1-\epsilon)}{3} \cdot \left(\frac{\epsilon}{1-\epsilon}\right)^2\right) = \exp\left(-\frac{\epsilon^2}{3(1-\epsilon)} \cdot \frac{7 \ln n}{\epsilon^2}\right)$$

which simplifies to:

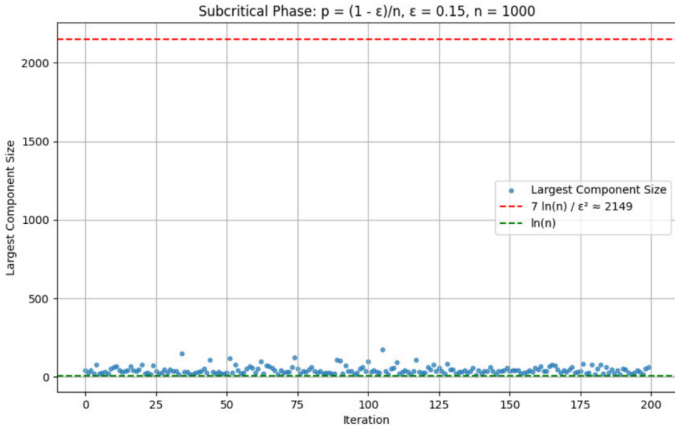
$$\Pr(S_I \geq k) \leq \exp\left(-\frac{7}{3(1-\epsilon)} \ln n\right) = n^{-\frac{7}{3(1-\epsilon)}}$$

This bounds the probability for a *single* interval of kn queries. However, the total number of queries possible in the entire duration of the algorithm is $\binom{n}{2}$. To ensure no such interval exists *anywhere* in the entire run, we apply the Union Bound over all $\binom{n}{2} - kn + 1$ possible starting positions:

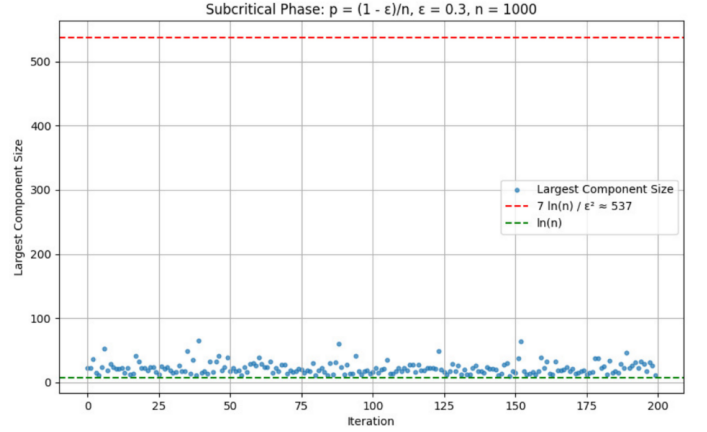
$$\Pr[\text{DFS finds } k \text{ edges in any } kn \text{ consecutive queries}] \leq \left(\binom{n}{2} - kn + 1 \right) \cdot n^{-\frac{7}{3(1-\epsilon)}} \leq n^{2-\frac{7}{3(1-\epsilon)}}$$

For any $\epsilon > 0$, the term $\frac{7}{3(1-\epsilon)}$ is strictly greater than 2. Therefore, the overall exponent is strictly negative. As $n \rightarrow \infty$, this probability vanishes entirely (inverse polynomial).

Our algorithm required an event to happen (finding k edges in atmost kn queries) that we just proved occurs with *very less* probability in the limit). This contradicts our assumption. Consequently, **whp**, the stack U must empty out long before the $(k + 1)$ -th vertex is ever discovered, proving that no component can exceed size $\frac{7}{\epsilon^2} \ln n$.



(a) Subcritical: $\epsilon = 0.15$



(b) Subcritical: $\epsilon = 0.3$

Figure 2: Fixing n to be 1000 and adding edges with different probability ≈ 200 times.

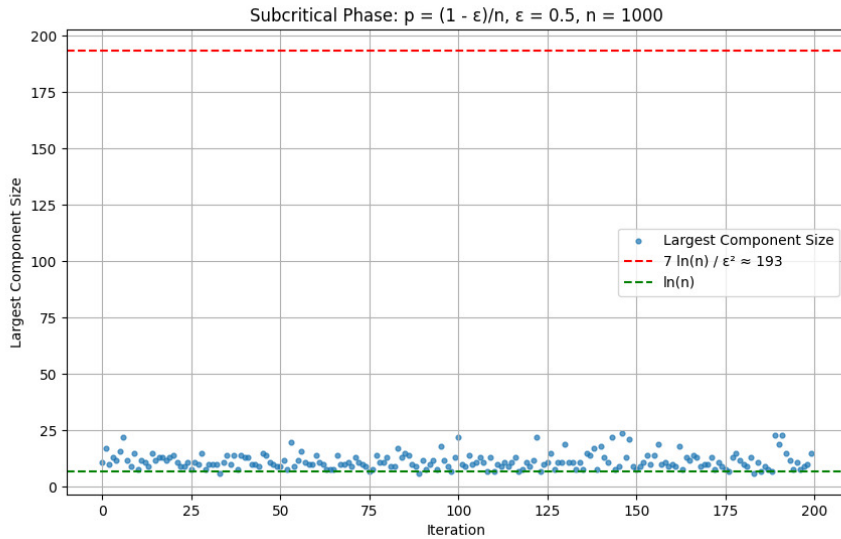


Figure 3: Subcritical: $\epsilon = 0.5$ for same n .

5 A bit to the right... $p \geq \frac{1+\epsilon}{n}$

We now cross the critical threshold, and all the logarithmic components seems to be *combine* to form linear sized components.

Theorem 5.1. *Let $\epsilon > 0$ be a sufficiently small constant, and let $G \sim G(n, p)$ with $p = \frac{1+\epsilon}{n}$. Then **whp**, G contains a connected component of size at least $\frac{\epsilon^2 n}{5}$, which constitutes a linear fraction $\Theta(n)$ of the entire graph.*

We again proceed by contradiction. Assume that G does *not* contain any connected component of size at least $k = \frac{\epsilon^2 n}{5}$.

We run our DFS algorithm. Because the stack U represents a single, continuous path of nodes within the specific component currently being explored, the size of U can never exceed the total size of that component. Therefore,

$$|U| < \frac{\epsilon^2 n}{5} \quad \text{at all times.}$$

Claim: After $N_0 = \frac{\epsilon n^2}{2}$ queries, $|U| \geq \frac{\epsilon^2 n}{5}$

These N_0 queries consist of some “yes” or “no” queries (X_t being 1 or 0), and number of “no” queries is lower bounded by $|S||T|$ (for every vertex v that had been added to S must have queried against each vertex in T to see if it can be *extended*). Now $|T| = n - |S| - |U|$ and since we assumed $|U| < \frac{\epsilon^2 n}{5}$ we know

$$N_0 \geq |S||T| = |S|(n - |S| - |U|) > |S| \left(n - |S| - \frac{\epsilon^2 n}{5} \right) \quad \text{call this } (\star)$$

The last expression is an inverted parabola with roots at 0 and $n - \epsilon^2 n/5 \sim n$, so for 0 to M it is increasing while for M to n it is decreasing, where $M \sim n/2$ for small enough ϵ .

Next question: **Can we get some bound on $|S|$? Upper bound? Lower bound?**

We would very much like $|S|$ to be less than $n/2$ so that we can proceed in the above inequality. Infact, an even stronger result holds!

Sub-Claim: $|S| < n/3$.

Suppose not, then $|S| > n/3$. There exists a moment when $|S| = n/3$. As $|T| = n - |S| - |U|$, *if* (and that’s a big *if*) we could prove $|U| < n/3$ then $|T| > n - n/3 - n/3 = n/3$ and then since $N_0 \geq |S||T| \geq n/3 \cdot n/3 = n^2/9 > N_0$, contradiction.

Sub-Claim: Let $p = \frac{1+\epsilon}{n}$. Let $N_0 = \frac{\epsilon n^2}{2}$. Then whp $\left| \sum_{i=1}^{N_0} X_i - \frac{\epsilon(1+\epsilon)n}{2} \right| \leq n^{2/3}$.

First, we calculate the expected value of our sum S .

$$\mu = \mathbb{E}[S] = N_0 p = \left(\frac{\epsilon n^2}{2} \right) \left(\frac{1+\epsilon}{n} \right) = \frac{\epsilon(1+\epsilon)n}{2}$$

To bound this deviation, we use chevyshev’s inequality, and for that we need to find the variance of S . For a Binomial distribution $B(N_0, p)$, the variance is:

$$\text{Var}(S) = N_0 p(1-p) = \mu(1-p) = \left(\frac{\epsilon \cdot (1+\epsilon)n}{2} \right) \cdot \left(\frac{n-1-\epsilon}{n} \right)$$

and

$$\Pr(|S - \mu| \geq t) \leq \frac{\text{Var}(S)}{t^2}$$

Substituting the values of $\text{Var}(S)$ and $t = n^{2/3}$:

$$\Pr(|S - \mu| \geq n^{2/3}) \leq \frac{\epsilon \cdot (1 + \epsilon) \cdot (n - 1 - \epsilon)}{2n^{4/3}} = \mathcal{O}(n^{-1/3})$$

for small enough constant ϵ . As the graph size $n \rightarrow \infty$, the fraction $\mathcal{O}(n^{-1/3})$ strictly approaches 0. Therefore, the probability that the sum deviates by more than $n^{2/3}$ vanishes entirely. Consequently, the complementary event that the sum is tightly bounded within $n^{2/3}$ of its expectation must hold (**whp**).

Recall that $|U| < 1 + \sum_{i=1}^t X_i$ and since sum of X_i is bounded in the region $[\mu - n^{2/3}, \mu + n^{2/3}]$, where $\mu = \mathcal{O}(n)$, $|U| < n/3$ holds. This proves our claim.

Now let's establish a lower bound on $|S|$ as well: Recall once again $|S \cup U| \geq \sum_{i=1}^t X_i$. Indeed,

$$|S| \geq |S \cup U| - |U| > \sum_{i=1}^t X_i - \frac{\epsilon^2 n}{5} \geq \frac{\epsilon(1 + \epsilon)n}{2} - n^{2/3} - \frac{\epsilon^2 n}{5} = -n^{2/3} + \frac{\epsilon n}{2} + \frac{3\epsilon^2 n}{10}$$

We are almost done! We earlier established that (see $(\star)$), $N_0 \geq |S| \cdot \left(n - |S| - \frac{\epsilon^2 n}{5}\right)$. Now since $|S| < n/3 < n/2$, the middle point of this downward parabola, we thus get (we can substitute $|S|$ by its lower-bound found above):

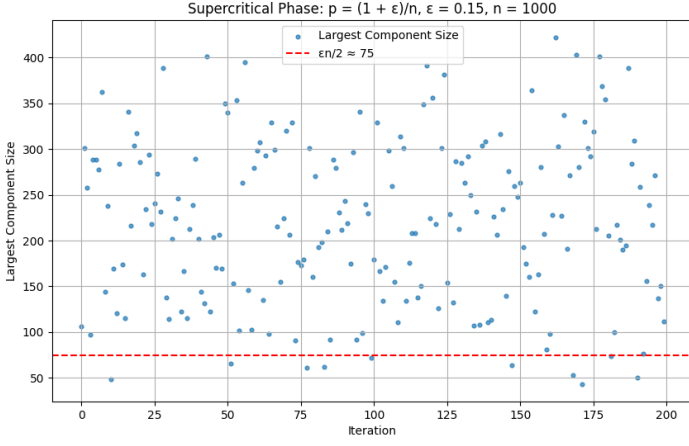
$$\begin{aligned} \frac{\epsilon n^2}{2} = N_0 &\geq |S| \left(n - |S| - \frac{\epsilon^2 n}{5}\right) \\ &\geq \left(\frac{\epsilon n}{2} + \frac{3\epsilon^2 n}{10} - n^{2/3}\right) \left(n - \frac{\epsilon n}{2} - \frac{\epsilon^2 n}{2} + n^{2/3}\right) \\ &= \frac{\epsilon n^2}{2} + \frac{\epsilon^2 n^2}{20} - \mathcal{O}(\epsilon^3) n^2 - \mathcal{O}(n^{5/3}) > \frac{\epsilon n^2}{2} \end{aligned}$$

For what values of ϵ , will this contradiction hold? In other words, for what values of ϵ will we stay outside the critical window?

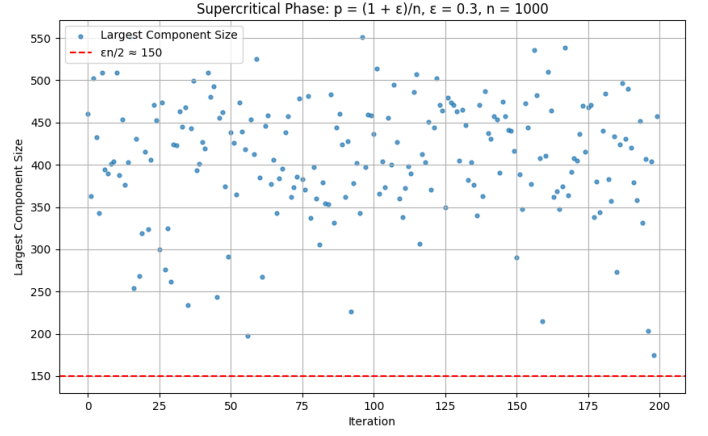
For our final strict inequality ($> \frac{\epsilon n^2}{2}$) to hold, the value of $(\frac{\epsilon^2 n^2}{20})$ must be large enough to strictly overpower both the negative terms. This requires:

$$\epsilon^2 n^2 \gg n^{5/3} \implies \epsilon^2 \gg n^{-1/3} \implies \epsilon \gg n^{-1/6}$$

If ϵ is a fixed positive constant, this condition is trivially satisfied as $n \rightarrow \infty$, and we are good, proving the existence of the Giant Component. True *theoretical* critical window comes when $\epsilon = \Theta(n^{-1/3})$ while our proof only holds till ϵ reaches $n^{-1/6}$. Turns out if we use chernoff bound instead of chevyshev in our proof, we can push this till $n^{-1/3} \ln^{1/3} n$. A proof of this (this was not included in the paper) can be found in the appendix of this report.



(a) Supercritical: $\epsilon = 0.15$



(b) Supercritical: $\epsilon = 0.3$

Figure 4: Fixing n to be 1000 and adding edges with different probability ≈ 200 times.

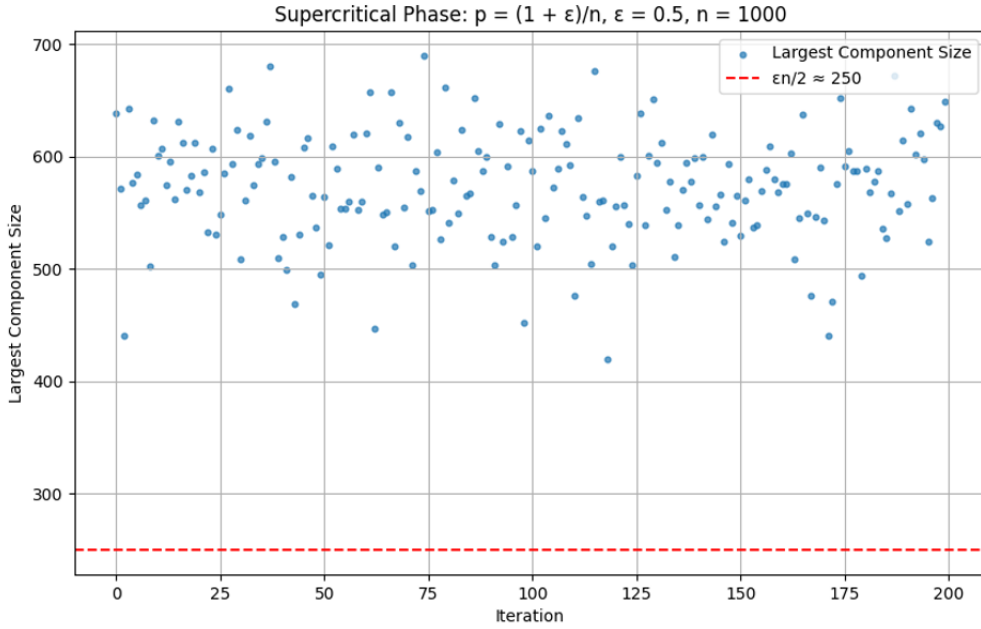


Figure 5: Supercritical: $\epsilon = 0.5$

Note that in previous proof, we actually showed that after N_0 queries, size of stack must be at least $\epsilon^2 n/5$. Since $|U|$ at anytime stores a path, we actually proved that a path of such length exists! We claim that we have a slightly stronger result when we specifically try to analyze for the *size of the component*.

Theorem 5.2. Let $p = \frac{1+\epsilon}{n}$, for $\epsilon > 0$ a small enough constant. Let $G \sim G(n, p)$. Then **whp** G has a connected component with at least $\frac{\epsilon n}{2}$ vertices.

Proof. The proof is very similar to the previous one. Assume N_0 as $\epsilon n^2/2$. It is easy to show that the following two property hold **whp** (using Chernoff):

1. $\sum_{i=1}^{n^{7/4}} X_i \leq n^{5/6}$
2. For every $n^{7/4} \leq t \leq N_0$: $\left| \sum_{i=1}^t X_i - (1 + \epsilon) \frac{t}{n} \right| \leq n^{2/3}$

Assume $\bar{X}$ satisfies these properties. We claim that after exactly N_0 queries, the DFS is still executing the same epoch (exploring the *same* connected component), and its size is already at least $\frac{\epsilon n}{2}$.

Using Property 2, at any moment $n^{7/4} \leq t \leq N_0$, we know that:

$$|S \cup U| \geq \sum_{i=1}^t X_i \geq (1 + \epsilon) \frac{t}{n} - n^{2/3}$$

Also, here too $|S| < n/3$ holds just like it did in the previous case. Suppose for the sake of contradiction that at some moment t in this interval, the stack U becomes completely empty. If U is empty, it means the current connected component is fully explored, and the algorithm has asked all queries between the set S and its complement T (with their answer being 0, specifically $X_i = 0$). Since the total queries asked so far is t , we have:

$$t \geq |S|(n - |S|)$$

Let $x = |S|$. Because $t \leq \frac{\epsilon n^2}{2}$, we know $x \approx \epsilon t/n \leq \epsilon^2 n/2$, which is strictly less than $n/2$ for small ϵ . Because the parabola $x(n - x)$ is monotonically increasing for $x \leq n/2$, we can substitute our lower bound for $|S \cup U|$ into the inequality (since $|U| = 0, |S \cup U| = |S|$):

$$t \geq \left((1 + \epsilon) \frac{t}{n} - n^{2/3} \right) \left(n - (1 + \epsilon) \frac{t}{n} + n^{2/3} \right)$$

Simplifying the expression we get:

$$t \geq (1 + \epsilon)t - n^{5/3} - (1 + \epsilon)^2 \frac{t^2}{n^2} + 2(1 + \epsilon)tn^{-1/3} - n^{4/3}$$

and then

$$t \geq (1 + \epsilon)t - (1 + \epsilon)^2 \frac{t^2}{n^2} - 2n^{5/3}$$

Subtract t from both sides:

$$0 \geq \epsilon t - (1 + \epsilon)^2 \frac{t^2}{n^2} - 2n^{5/3}$$

$$0 \geq \epsilon - (1 + \epsilon)^2 \frac{t}{n^2} - \frac{2n^{5/3}}{t}$$

We observe this inequality at its limits within the interval $t \in [n^{7/4}, \frac{\epsilon n^2}{2}]$:

- Because $t \leq \frac{\epsilon n^2}{2}$, the middle term is strictly bounded: $(1 + \epsilon)^2 \frac{t}{n^2} \leq (1 + \epsilon)^2 \frac{\epsilon}{2} \approx \frac{\epsilon}{2}$.
- Because $t \geq n^{7/4}$, the final term vanishes asymptotically: $\frac{2n^{5/3}}{t} \leq \frac{2n^{5/3}}{n^{7/4}} = 2n^{-1/12} \rightarrow 0$.

Substitute them back we get:

$$0 \geq \epsilon - \frac{\epsilon}{2} - 0 \implies 0 \geq \frac{\epsilon}{2}$$

A contradiction (as $\epsilon > 0$). This contradiction proves our assumption was false: the stack U **cannot** become empty at any point in this interval.

Consequently, at $t = N_0$, the algorithm is still exploring the very same connected component. The size of this component at time N_0 is at least $|S \cup U|$, which by Property 2 is:

$$|S \cup U| \geq (1 + \epsilon) \frac{N_0}{n} - n^{2/3} = (1 + \epsilon) \frac{\epsilon n}{2} - n^{2/3} = \frac{\epsilon n}{2} + \frac{\epsilon^2 n}{2} - n^{2/3}$$

For sufficiently large n , this value is strictly greater than $\frac{\epsilon n}{2}$. Therefore, **whp**, a giant connected component of at least size $\frac{\epsilon n}{2}$ exists. $\square$

6 Critical Region: $p = 1/n$

At the exact phase transition threshold ($p = 1/n$, corresponding to), the graph undergoes its most chaotic shift. It is known that the size of the largest connected component scales as $\Theta(n^{2/3})$.

Rigorously proving this scaling bound is notoriously difficult, requiring advanced probabilistic techniques such as martingale theory and complex branching processes, which fall beyond the scope of this report. Therefore, rather than providing a formal mathematical derivation for the critical region, we demonstrate this empirically.

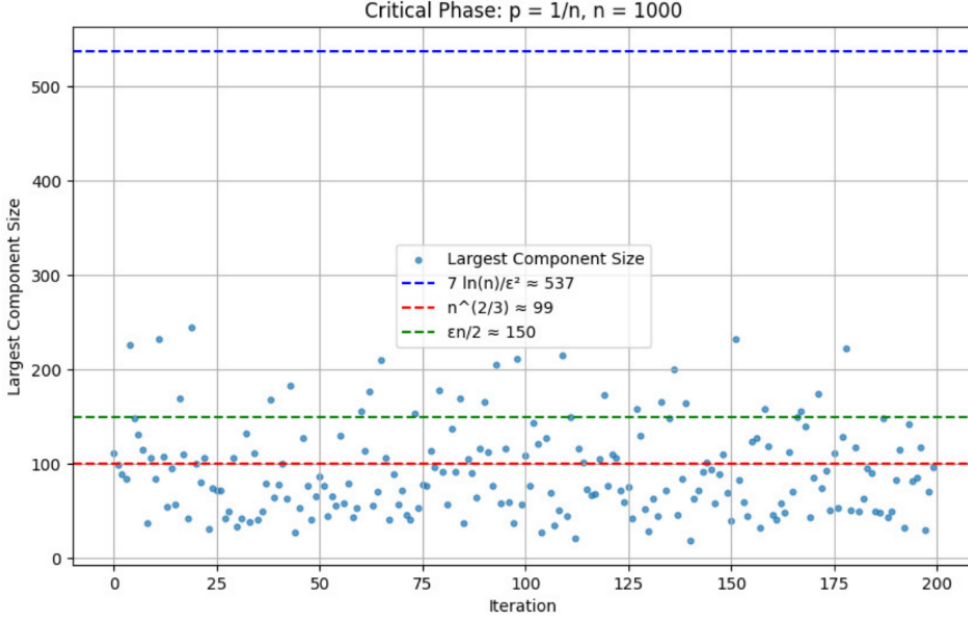


Figure 6: Critical: $p = 1/n$.

7 When does the graph become connected?

Even after the emergence of the Giant Component, the graph is not entirely connected. The problem is that there are still a lot of tiny components left behind. As we continue to increase the edge probability p , the Giant linear sized components combine together (even the smaller ones too) and the graph becomes connected.

We show that this transition occurs at exactly $p = \frac{\ln n}{n}$. (We first actually proved that with $p = c \ln n/n$, the graph becomes connected whp, but when we delve deeper we came across this much more seemingly non-trivial statement.)

Theorem 7.1. *Let $p = \frac{\ln n + c_n}{n}$, where c_n is a sequence such that $|c_n| = o(n^{1/2})$. (This is something we have chosen for the proof.) Let $p = \frac{\ln n + c_n}{n}$.*

1. *If $c_n \rightarrow \infty$ as $n \rightarrow \infty$, then **whp** $G(n, p)$ is connected.*
2. *If $c_n \rightarrow -\infty$ as $n \rightarrow \infty$, then **whp** $G(n, p)$ is disconnected.*

We split the proof into three parts. First, we analyze the components of size $k = 1$ in both regimes. Let $X = \sum_{v \in V} I_v$ be the total number of isolated vertices, where I_v is an indicator variable that equals 1 if vertex v has degree 0, and 0 otherwise.

7.1 The Lower Bound: $c_n \rightarrow -\infty$

If we do not add enough edges, isolated vertices will survive **whp**, ensuring the graph remains disconnected. We show this using Chebyshev's Inequality.

Proof. First, we calculate the expected number of isolated vertices, $\mu = \mathbb{E}[X]$. For a specific vertex to be isolated, all $n - 1$ of its potential edges must fail to exist:

$$\mathbb{E}[I_v] = (1 - p)^{n-1}$$

By LOE (Linearity of Expectation):

$$\mu = n(1 - p)^{n-1}$$

First, note that as $n \rightarrow \infty$, $p \rightarrow 0$. To analyze this as $n \rightarrow \infty$, we take the natural logarithm and apply the Taylor expansion $\ln(1 - p) = -p - \frac{p^2}{2} - \mathcal{O}(p^3)$:

$$\begin{aligned} \ln(\mu) &= \ln n + (n - 1) \ln(1 - p) \\ &\approx \ln n - n \left(\frac{\ln n + c_n}{n} \right) - \frac{n}{2} \left(\frac{\ln n + c_n}{n} \right)^2 \\ &= -c_n - \frac{(\ln n + c_n)^2}{2n} \end{aligned}$$

Assuming c_n grows slower than $n^{1/2}$ (i.e., $|c_n| = o(n^{1/2})$), the latter error term vanishes exactly to 0. Thus, the asymptotically: $\ln(\mu) \sim -c_n \implies \mu \sim e^{-c_n}$. Because $c_n \rightarrow -\infty$, the expected number of isolated vertices $\mu \rightarrow \infty$.

The variance of X is $\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$. By expanding $\mathbb{E}[X^2] = \mathbb{E}[(\sum I_v)^2] = \sum \mathbb{E}[I_v] + \sum_{u \neq v} \mathbb{E}[I_u I_v]$. Now for $\mathbb{E}[I_u I_v]$, note that the probability for two distinct vertices u and v to both have $\deg(u) = \deg(v) = 0$, we require that neither connects to the remaining $n - 2$ vertices, and the edge between them does not exist as well. Indeed,

$$\mathbb{E}[I_u I_v] = (1 - p)^{2n-3}$$

Next, we substitute this back to find the variance. Factoring out $\mu^2 = n^2(1 - p)^{2n-2}$ from the middle term gives:

$$\begin{aligned} \text{Var}(X) &= \mu + n(n - 1)(1 - p)^{2n-3} - \mu^2 \\ &= \mu + \mu^2 \left(\frac{n - 1}{n} (1 - p)^{-1} - 1 \right) \end{aligned}$$

Using the Taylor expansion $(1 - p)^{-1} \approx 1 + p$ for small p :

$$\begin{aligned} \text{Var}(X) &\approx \mu + \mu^2 \left[\left(1 - \frac{1}{n} \right) (1 + p) - 1 \right] \\ &\approx \mu + \mu^2 \left(p - \frac{1}{n} \right) \approx \mu + \mu^2 p \end{aligned}$$

Applying Chebyshev's Inequality to bound the probability that $X = 0$ (meaning no isolated vertices exist):

$$\Pr(X = 0) \leq \Pr(|X - \mu| \geq \mu) \leq \frac{\text{Var}(X)}{\mu^2} \approx \frac{\mu + \mu^2 p}{\mu^2} = \frac{1}{\mu} + p$$

Since $\mu \rightarrow \infty$ (because $c_n \rightarrow -\infty$), the fraction $\frac{1}{\mu} \rightarrow 0$. Furthermore, $p = \frac{\ln n + c_n}{n} \rightarrow 0$. Therefore, the entire sum approaches exactly 0. Consequently, **whp**, $X > 0$. Isolated vertices exist, and the graph is disconnected. $\square$

7.2 The Upper Bound: $c_n \rightarrow \infty$

We now prove that if we push slightly past the threshold, the graph becomes connected

Proof. We return to our expectation from the previous section:

$$\mu = \mathbb{E}[X] \approx e^{-c_n}$$

Because $c_n \rightarrow \infty$, the expected number of isolated vertices $\mu \rightarrow 0$.

Applying Markov's Inequality, the probability of finding even a single isolated vertex is:

$$\Pr(X \geq 1) \leq \mathbb{E}[X] \rightarrow 0$$

Therefore, **whp**, the graph has absolutely no isolated vertices. But what about components of size 2 or 3 or any size $2 \leq k \leq n/2$ for that matter? We require a bit more algebra! For the graph to become fully connected, we require that absolutely no components of any size $2 \leq k \leq n/2$ exist. (Note that we do not need to look if some component of size $k > n/2$ exists or not.) Let X_k be the number of connected, isolated components of size k in $G(n, p)$. For a set of k vertices to form such a component, we know that two conditions must be met:

1. They must form at least a spanning tree among themselves. By Cayley's formula, there are k^{k-2} such possible trees, and each requires $k - 1$ specific edges to exist.
2. They must be completely severed from the remaining $n - k$ vertices. This requires exactly $k(n - k)$ potential edges to not exist.

Using the union bound over all $\binom{n}{k}$ possible subsets of size k , the expected number of such components is bounded by:

$$\mathbb{E}[X_k] \leq \binom{n}{k} k^{k-2} p^{k-1} (1 - p)^{k(n-k)}$$

Applying $\binom{n}{k} \leq \left(\frac{en}{k}\right)^k$ and $(1 - p) \leq e^{-p}$, we get:

$$\mathbb{E}[X_k] \leq \left(\frac{en}{k}\right)^k k^{k-2} p^{k-1} \exp(-pk(n - k))$$

It turns out that we must deal with this in two different ranges of k differently.

Case 1: Small Components ($2 \leq k \leq n^{1/4}$)²

We expand the exponential term algebraically: $\exp(-pk(n - k)) = \exp(-pkn + pk^2)$. Because $k \leq n^{1/4}$ and $cn = o(n^{1/2})$:

$$pk^2 = \left(\frac{\ln n + c_n}{n}\right) k^2 \leq \frac{n^{1/2}(\ln n + c_n)}{n} \rightarrow 0$$

Thus, for sufficiently large n , $\exp(pk^2) \leq 2$. We substitute this exact bound and $p = \frac{\ln n + c_n}{n}$:

$$\begin{aligned} \mathbb{E}[X_k] &\leq \left(\frac{en}{k}\right)^k k^{k-2} \left(\frac{\ln n + c_n}{n}\right)^{k-1} e^{-pkn} \cdot 2 \\ &= 2 \frac{e^k n^k (\ln n + c_n)^{k-1}}{k^2 n^{k-1}} n^{-k} e^{-kc_n} \\ &= \frac{2e^k}{k^2} n^{1-k} (\ln n + c_n)^{k-1} e^{-kc_n} \end{aligned}$$

²Note that we could have taken any x , and analyzed everything in $2 \leq k \leq x$ and $x \leq k \leq n/2$ as long as $x = n^{1/t}$ for $t > 2$, like $n^{1/3}, n^{1/4}$. This can be noted in the proof of the larger components part. we believe this can be generalized a bit more but we couldn't do so.

For $k \geq 2$, the term $n^{1-k} \leq n^{-1}$. Since the ratio between consecutive terms in this sequence is something like $\mathcal{O}\left(\frac{\ln n + c_n}{n}\right) \rightarrow 0$, the sum is strictly dominated by its first term ($k = 2$):

$$\sum_{k=2}^{n^{1/4}} \mathbb{E}[X_k] = \mathcal{O}(\mathbb{E}[X_2]) = \mathcal{O}\left(\frac{\ln n + c_n}{n} e^{-2c_n}\right)$$

Because n grows vastly faster than $\ln n$, this entire sum mathematically vanishes to exactly 0.

Case 2: Large Components ($n^{1/4} < k \leq n/2$)

For these larger components, note that $k(n - k) \geq \frac{kn}{2}$. We substitute this into our original bound:

$$\begin{aligned} \mathbb{E}[X_k] &\leq \left(\frac{en}{k}\right)^k k^{k-2} \left(\frac{\ln n + c_n}{n}\right)^{k-1} \exp\left(-pk\frac{n}{2}\right) \\ &\leq (en)^k \frac{(\ln n + c_n)^{k-1}}{n^{k-1}} n^{-k/2} e^{-kc_n/2} \\ &\leq n \cdot \left(\frac{e(\ln n + c_n)}{n^{1/2} e^{c_n/2}}\right)^k \end{aligned}$$

Let B be the base of this exponential term: $B = \frac{e(\ln n + c_n)}{\sqrt{n} e^{c_n/2}}$. Because $\sqrt{n}$ dominates the logarithm, $B \rightarrow 0$ as $n \rightarrow \infty$. We sum this strict geometric series over the remaining interval:

$$\sum_{k=n^{1/4}}^{n/2} n \cdot B^k \leq n \sum_{j=n^{1/4}}^{\infty} B^j = n \left(\frac{B^{n^{1/4}}}{1 - B}\right) \approx n B^{n^{1/4}}$$

Because $B \rightarrow 0$, raising it to the power of $n^{1/4}$ ³ causes the whole value to tend to 0. Therefore, $\sum E[X_k]$ for $2 \leq k \leq n/2$ tends to 0.

Because the expectation sum drops to 0 across both intervals, it is guaranteed that **whp**, absolutely no components of size $2 \leq k \leq n/2$ exist. $\square$

Digression, a bit: It must be noted that the above proof works for *constant* c too (say if $c_n = \ln n$, then we can actually show that sum of all $\mathbb{E}[X_k]$ in $[1, n/2]$ is 0 without involving cayley's, we can trivially upper bound it by $\binom{n}{2}(1-p)^{k(n-k)}$ and then summing all of them up.) The main crux of the above proof was that no components of size *at least* 2 exists (even if c was constant). All we now know is that whp, we can say only isolated vertices exist as for constant c , $\mathbb{E}[X_1] \leq n(1-p)^{n-1}$, from the section 6.1, we can see that $\ln(\mathbb{E}[X_1]) \approx -c_n - \frac{(\ln n + c_n)^2}{2n}$ which implies $\mathbb{E}[X_1] \sim e^{-c_n}$. Since c_n is constant, this doesn't necessarily mean that $\mathbb{E}[X_1]$ is 0, there might exist isolated vertices! See Table 1.

³If we had taken the range till $n^{1/x}$, $x \geq 2$, all we need c_n to be is $o(n^{1-2/x})$ so that the proof for the small components doesn't break. All part 2 requires is a exponent such that $nB^{n^{1/x}}$ tends to 0.

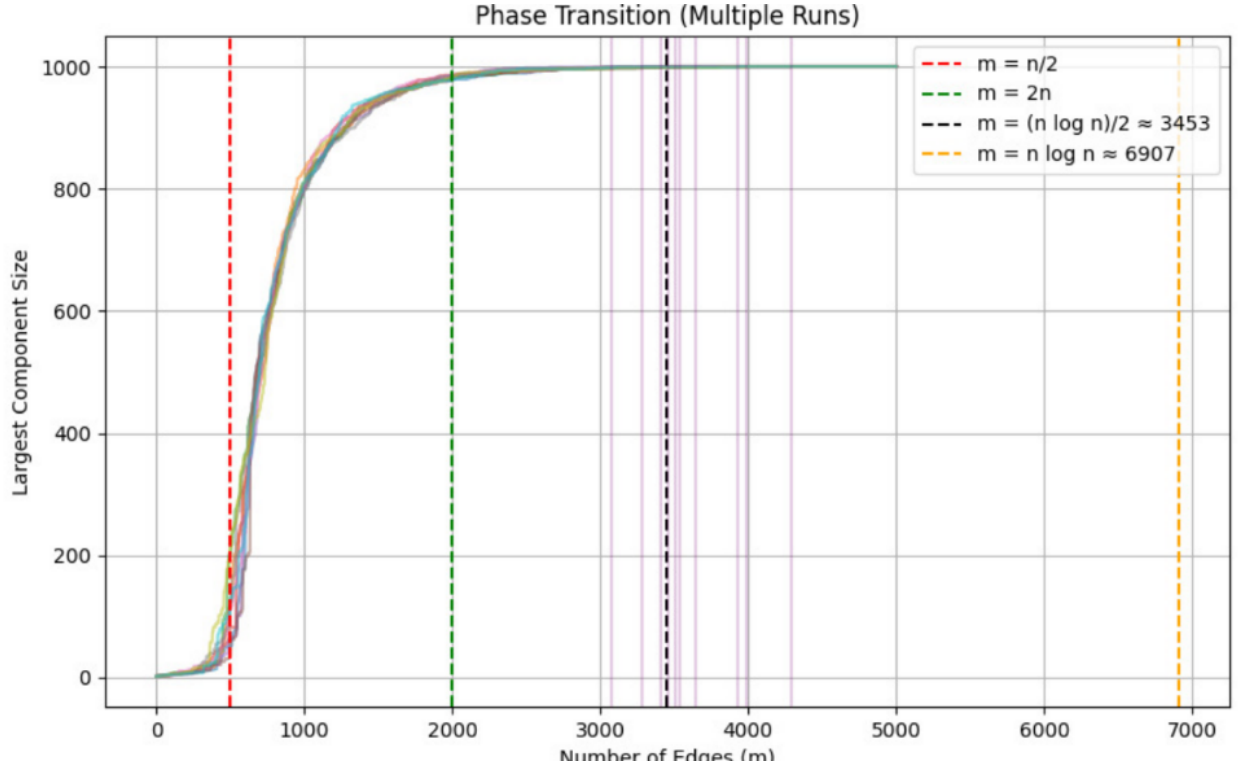


Figure 7: Here $n = 1000$, and $p = \ln n/n$, sometimes graph is connected (represented by pink lines being at left of black line, which represents that the graph is connected) or disconnected (represented by pink lines being at right of black line) marking the exact transition point, while at $p = 2 \ln n/n$ edges, graph is almost always connected.

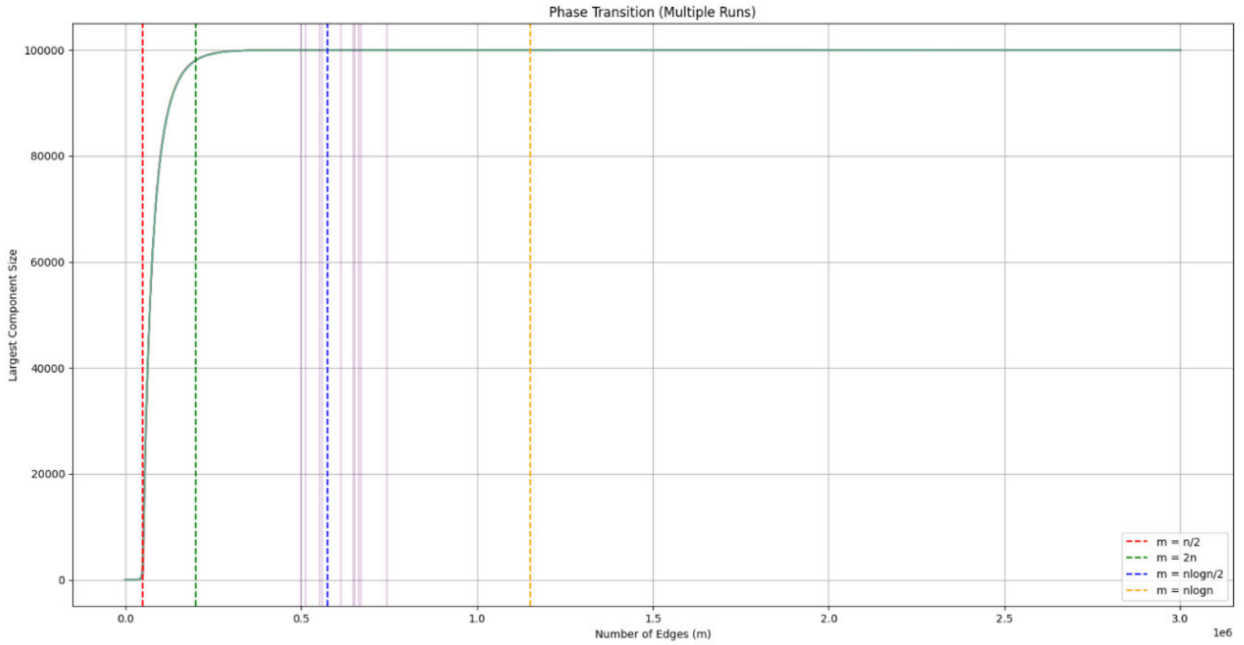


Figure 8: Here $n = 10^5$.

Edges	1	2	995	996	997	998	999	1000
3403	1.06	0.02	0.02	0.02	0.10	0.18	0.26	0.42
3413	1.06	0.02	0.02	0.02	0.10	0.18	0.26	0.42
3423	1.04	0.02	0.02	0.02	0.08	0.20	0.26	0.42
3433	1.02	0.02	0.02	0.02	0.06	0.22	0.26	0.42
3443	1.00	0.02	0.02	0.02	0.06	0.22	0.24	0.44
3453	0.98	0.02	0.02	0.02	0.06	0.20	0.26	0.44
3463	0.94	0.02	0.02	0.02	0.06	0.18	0.26	0.46
3473	0.94	–	0.02	0.02	0.04	0.18	0.28	0.46
3483	0.90	–	0.02	–	0.06	0.18	0.26	0.48
3493	0.88	–	0.02	–	0.06	0.16	0.28	0.48
3503	0.86	–	0.02	–	0.06	0.14	0.30	0.48

Table 1: Average number of components of given sizes near the connectivity threshold $m \approx \frac{n \log n}{2}$. It can be seen that almost all of the components are of size 1.

8 Conclusion

In this report, we explored the evolution of the Erdős-Rényi random graph, $G(n, p)$. By applying Chebyshev and Chernoff bounds alongside the standard DFS algorithm, we saw how the graph evolved through its distinct critical regions. It begins as a collection of small components, undergoes a sudden phase transition to birth a massive "Giant Component," and finally achieves total connectivity when the edge probability p crosses the $\frac{\ln n}{n}$ threshold.

A Pushing ϵ to the critical window when $p \geq \frac{1+\epsilon}{n}$

In Section 5, we rigorously established the emergence of the Giant Component. However, by utilizing Chebyshev's inequality to bound the variance of our DFS queries, our contradiction mathematically broke down when $\epsilon \sim n^{-1/6}$.

Because our random variables are independent Bernoulli trials, their sum is actually far more concentrated than Chebyshev's variance-based bound suggests. By applying the exponential Chernoff bounds, we can drastically shrink the probabilistic error term and push the validity of our DFS proof significantly closer to the true critical window of $\epsilon = \Theta(n^{-1/3})$.

We restate our sub-claim (regarding concentration of $\sum_{i=1}^{N_0} X_i$ about its mean), replacing the $n^{2/3}$ deviation with a tighter limit.

Lemma A.1. *Let $p = \frac{1+\epsilon}{n}$ and $N_0 = \frac{\epsilon n^2}{2}$. Let $S = \sum_{i=1}^{N_0} X_i$. For any sufficiently large n , **whp**:*

$$|S - \mu| \leq \sqrt{3\mu \ln n}$$

Proof. The expected value remains $\mu = \frac{\epsilon(1+\epsilon)n}{2} \approx \frac{\epsilon n}{2}$. We apply the standard two-sided Chernoff bound for an absolute error margin $t = \sqrt{3\mu \ln n}$:

$$\Pr(|S - \mu| \geq t) \leq 2 \exp\left(-\frac{t^2}{3\mu}\right)$$

Substituting this t into the bound yields:

$$\Pr(|S - \mu| \geq \sqrt{3\mu \ln n}) \leq 2 \exp(-\ln n) = \frac{2}{n}$$

As $n \rightarrow \infty$, $2/n \rightarrow 0$. Therefore, **whp**, the sum Z is restricted to a much tighter window of width $\sqrt{3\mu \ln n}$. $\square$

A.1 Re-evaluating the Contradiction

We now substitute this tighter error bound back into our DFS combinatorial bottleneck. Recall that our contradiction relied on substituting the lower bound of $|S|$ into the query equation $N_0 \geq |S| \left(n - |S| - \frac{\epsilon^2 n}{5} \right)$.

Using our new lemma, the lower bound for $|S|$ becomes:

$$|S| > \sum_{i=1}^{N_0} X_i - \frac{\epsilon^2 n}{5} \geq \frac{\epsilon n}{2} + \frac{3\epsilon^2 n}{10} - \sqrt{\frac{3}{2}\epsilon n \ln n}$$

When we expand the term $|S|(n - |S| - \frac{\epsilon^2 n}{5})$ with this new lower bound

$$\begin{aligned} N_0 &\geq |S| \left(n - |S| - \frac{\epsilon^2 n}{5} \right) \\ &\geq \left(\frac{\epsilon n}{2} + \frac{3\epsilon^2 n}{10} - \sqrt{\frac{3}{2}\epsilon n \ln n} \right) \left(n - \frac{\epsilon n}{2} - \frac{\epsilon^2 n}{2} + \sqrt{\frac{3}{2}\epsilon n \ln n} \right) \\ &= \frac{\epsilon n^2}{2} + \frac{\epsilon^2 n^2}{20} - \mathcal{O}(\epsilon^3) n^2 - \sqrt{\frac{3}{2}\epsilon n \ln n} \left(n - \epsilon n - \frac{4\epsilon^2 n}{5} \right) - \frac{3}{2}\epsilon n \ln n \end{aligned}$$

The final squared term $\frac{3}{2}\epsilon n \ln n$ scales as $\mathcal{O}(n \ln n)$ and is asymptotically negligible. For small ϵ , the coefficient in the brackets $(n - \epsilon n - \frac{4\epsilon^2 n}{5}) \approx n$.

Therefore, for the overall expression to strictly exceed our limit $N_0 = \frac{\epsilon n^2}{2}$, the term (which $\epsilon^2 n^2$ must dominate) in our expansion is exactly $n\sqrt{\frac{3}{2}\epsilon n \ln n} = \Theta(\epsilon^{1/2} n^{3/2} \ln^{1/2} n)$. Therefore,

$$\frac{\epsilon^2 n^2}{20} \gg \epsilon^{1/2} n^{3/2} \ln^{1/2} n$$

Solving for ϵ :

$$\epsilon^{3/2} \gg n^{-1/2} \ln^{1/2} n \implies \epsilon \gg n^{-1/3} \ln^{1/3} n$$

Therefore, by using much stronger Chernoff bound, we mathematically proved that the DFS contradiction survives right up to the boundary of $\epsilon \gg n^{-1/3} \ln^{1/3} n$ which is close to the theoretical critical window of $\epsilon = \Theta(n^{-1/3})$.

References

- [1] P. Erdős and A. Rényi, *On random graphs I*. Publicationes Mathematicae Debrecen, **6** (1959), 290–297.
- [2] M. Krivelevich and B. Sudakov, *The phase transition in random graphs: a simple proof*. Random Structures & Algorithms, **43**(2) (2013), 131–138.
- [3] B. Bollobás, *Random Graphs*, 2nd ed. Cambridge University Press, 2001.